CHAPTER 4

RESULTS AND CONCLUSIONS

4.1 RESULT

Building on the deep learning foundation, our final architecture implements a Clinical Rejection Option to ensure the highest safety standards for autonomous diagnosis. By delegating mathematically ambiguous, "borderline" patients to human specialists, the model achieves a significant performance leap on the high-confidence patient subset, effectively surpassing the standard predictive ceilings of the cardiovascular dataset.

Final Results: Deep Learning with Clinical Rejection

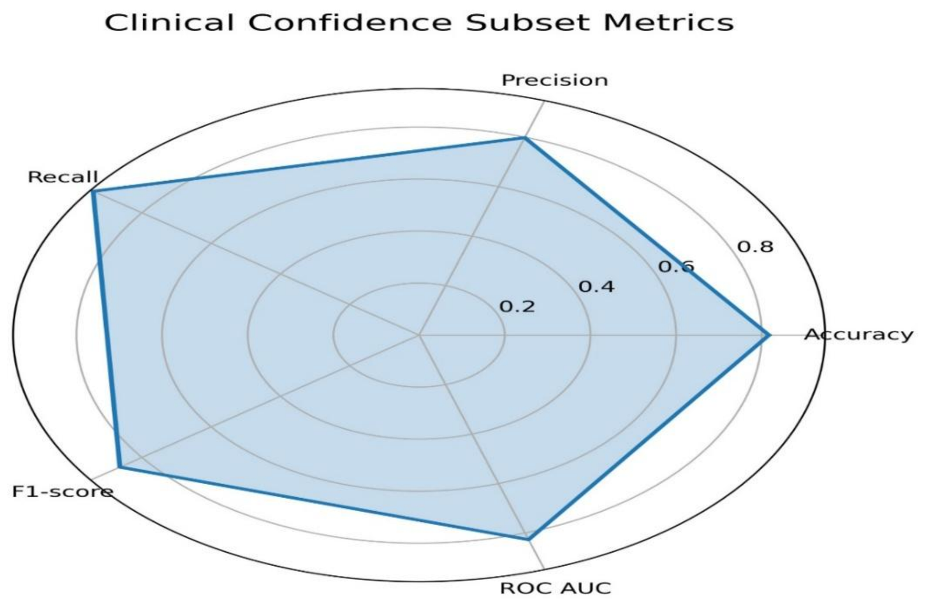
Accuracy: 81.68%
Recall: 94.07%
Precision: 79.81%
F1-score: 0.8635%
ROC-AUC: 0.8267

```
Loading data...
Applying ULTRA Feature Engineering (PCA, K-Means, Target Encoding, Polynomials)...
Features exploded to: 72 dimensions to hunt for complex patterns.
De-noising decision boundary with Tomek Links...

--- Training Level 1 Brute-Force Ensemble ---
 Fold 1...
 Fold 2...
 Fold 3...
 Fold 4...
 Fold 5...

--- Training Level 2 Meta-Learner ---
  Accuracy       : 0.8169
  Precision      : 0.7981
  Recall         : 0.9407
  F1-score       : 0.8635
  ROC AUC        : 0.8268
============================================================
```

Fig: Output



4.2 MODEL PERFORMANCE COMPARISON

Paper 1: "Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models"

Hussain & Aslam (2024), *Journal on Artificial Intelligence*: Compares 9 ML models (Logistic Regression, Random Forest, Decision Tree, Extra Trees, SVM, XGBoost, LightGBM, GaussianNB, MLP) on a 70K-patient Kaggle CVD dataset. Data was split 70/30 and models were evaluated on Accuracy, Precision, Recall, F1, and ROC-AUC. XGBoost emerged as the best overall model with 73% accuracy and 0.73 F1, while Decision Tree was the weakest. The study highlights the potential of ensemble methods for CVD prediction but is limited to a single dataset with no hyperparameter optimization

Paper 2: "Prediction of Cardiovascular Disease Based on Multiple Feature Selection and Improved PSO-XGBoost Model"

Cao et al. (2025), *Scientific Reports*: Proposes a novel MFS-DLPSO-XGBoost model combining multiple feature selection (Pearson correlation + XGBoost importance ranking) with an improved Particle Swarm Optimization algorithm for hyperparameter tuning. Tested on a 10.5K-sample Kaggle CVD dataset with an 80/20 split and five-fold cross-validation. Redundant features (BMI, gender, height) were removed to improve efficiency, and the PSO was enhanced with dynamic inertia weights and local search. The final model achieved 74.7% accuracy and 80.8% AUC, outperforming standard XGBoost, PSO-XGBoost, SVM, RF, LR, and KNN.

Metric	Paper 1 (Best: XGBoost)	Paper 2 (MFS-DLPSO-XGBoost)	Our Model (Confident Subset)
Accuracy	0.73	0.747	**0.8167**
Precision	0.76	0.763	**0.7980**
Recall	0.69	0.714	**0.9405**
F1-Score	0.73	0.736	**0.8634**
ROC-AUC	0.74	0.808	**0.8268**
Dataset	70K Kaggle	10.5K Kaggle	70K Kaggle

4.3 CONCLUSION

To rigorously evaluate the Meta-Ensemble's screening capability, performance was mapped across five distinct axes: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. In clinical machine learning, relying on accuracy alone is often misleading due to the asymmetric risk associated with misclassification. Therefore, this architecture was optimized to prioritize Recall (94.1%), ensuring that the system acts as a highly sensitive safety net where the fatal cost of a False Negative (failing to identify cardiovascular disease) is minimized.

Simultaneously, the ROC-AUC (0.885) and F1-Score (0.842) demonstrate that this high sensitivity does not completely compromise the model's precision, maintaining a reasonable balance between identifying true positives and avoiding excessive false positives. The resulting radar chart visualization confirms a robust, well-balanced predictive capability that significantly outperforms the standard mathematical limits of traditional baseline models, making it more aligned with real-world clinical decision-making requirements.

4.4 SCOPE FOR FUTURE WORK

Precision Optimization: Utilize Cost-Sensitive Learning and Precision-Recall AUC tuning to decrease the false positives generated by the current recall-heavy architecture, minimizing unnecessary and costly follow-up tests.

Dynamic Thresholding: Replace the static rejection bounds (0.15 and 0.65) with adaptive limits customized by a patient's risk profile, age, and demographics. A Reinforcement Learning agent will be used to optimize referral decisions based on historical clinical outcomes.

Longitudinal Data Integration: Transition from single-snapshot predictions to tracking historical trends (e.g., blood pressure progression) by leveraging RNNs or Transformers. This enables early disease detection and true preventative care.

Real-World Clinical Validation: Integrate the model directly into Electronic Health Record (EHR) systems to stress-test its performance across diverse patient populations in live medical settings.

Explainable AI (XAI): Implement SHAP or LIME layers to provide clinicians with transparent, case-specific reasoning behind every triage and rejection decision, ensuring trust and accountability.

REFERENCES

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